MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)
MG3DSCMAT204-Essentials of Financial Mathematics

Duration: 2 hrs Maximum Marks: 70

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Answer any five questions. Each question carries 2 marks

1. Define annual percentage rate (APR). [K][1]
2. Find the value in 4 years' time, of \$10,000 invested at 5% interest compounded annually. [U][1]
3. Find the dimension of the following system of equations
   $2x + 5y = 9$
   $7x - 4y = 2$
   $3x + 8y = 7$ [U][2]
4. Define break-even analysis. [A][2]
5. Define Extreme point. [A][3]
6. Define feasible region [E][3]
7. Find the critical points of $f(x) = 3x^3 - 54x^2 + 288x - 22$. [An][4]
8. Define marginal revenue in economics. [A][4]

Part B
Answer any five questions. Each question carries 6 marks

9. Determine the annual percentage rate of interest of a deposit account that has a nominal rate of 8% compounded monthly. [K][1]
10. A principal of \$100000 invested at 12% interest for 1 year. Determine the future value if the interest is compounded a) Annualy b) Semi annualy c) Quarterly . [An][1]
11. Solve the following system of equations graphically.
    $x - 2y = -2$
    $6x + 3y = 33$ [U][2]

12. Find the equilibrium price P_e and quantity Q_e for the following market
Supply : $P = 3Q + 10$
Demand : $P = \frac{-1}{2}Q + 80$. [An][2]

13. Solve the following system of equations graphically by finding the slope-intercept forms.
$12x + 3y = 21$
$-2x + 4y = -8$ [A][2]

14. Use graphical method, Maximize $C = 50x_1 + 30x_2$
Subject to the constraints
$2x_1 + x_2 \leq 14$
$5x_1 + 5x_2 \leq 40$
$x_1 + 3x_2 \leq 18$, $x_1, x_2 \geq 0$. [A][3]

15. Find the critical values and check whether it is relative maximum or minimum of the followings
a) $y = 9x^2 + 126x - 74$
b) $y = -5x^2 + 90x - 73$ [A][4]

16. Show that the marginal revenue must equal marginal cost at the profit-maximizing level of output. [A][4]

Part C
Answer any three questions. Each question carries 10 marks

17. A project requiring an initial outlay of $\$15,000$ is guaranteed to produce a return of $\$20,000$ in 3 years' time. Use the

    (a) Net present value

    (b) International rate of return

    methods to decide whether this investment is worthwhile if the prevailing market rate is 5% compound annually. [A][1]

18. Find the break-even point for the firms with the following linear total revenue $R(x)$ and total cost $C(x)$ functions by

    (a) finding the profit $\pi(x)$ function, setting it equal to zero and solving for x

    (b) setting $R(x) = C(x)$ and solving for x

    $R(x) = 55x$
    $C(x) = 30x + 250$ [An][2]

19. Solve the following system of equations
$2x + 5y = 52$
$3x - 4y = -14$ using

    (a) Elimination method

    (b) substitution method [An][2]

20. Find the optimal solution of the following maximization problem ,
Maximize $10x + 8y$
Subject to $6x + 2y \leq 36$
$3x + 5y \leq 30$
$x + 4y \leq 20$
$x, y \geq 0$ [An][3]

21. Find the marginal and average functions for each of the following total functions.Evaluate them at $Q = 2$ and $Q = 4$.

    (a) $TC = Q^2 + 9Q + 16$.

    (b) $TR = 24Q - Q^2$

    (c) $\pi = -Q^2 + 75Q - 12$. [A][4]

22. Optimize the following polynomial functions.

    (a) $y = 2x^4 - 8x^3 - 40x^2 + 79$

    (b) $y = -5x^4 + 20x^3 + 280x^2 - 19$.

    (c) $y = (7 - 2x)^4$. [A][4]